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Approximation Methods Graded Assignment

Graded Assignment • 10 min

Due Nov 13, 11:59 PM IST

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. **True** or **False**: The Inverse Power Method applied to the matrix $A = \begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix}$ and vector $\mathbf{x}_0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ converges. 1 / 1 point

   - ○ True
   - ◉ False

   Nice work

2. **True** or **False**: The Power Method applied to the matrix $A = \begin{bmatrix} 1 & 3 \\ 0 & -1 \end{bmatrix}$ and vector $\mathbf{x}_0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ converges. 1 / 1 point

   - ○ True
   - ◉ False

   Nice work

3. **True** or **False**: The Power Method applied to the matrix $A = \begin{bmatrix} 1 & 3 \\ 1 & -1 \end{bmatrix}$ and vector $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ converges. 1 / 1 point

   - ○ True
   - ◉ False

   Nice work

4. **True** or **False**: Suppose A is a 5×5 matrix having eigenvalues $-4, -2, 1, 4, \text{ and } 7$. Then $(A + I_5)^{-1}$ has dominant eigenvalue $\lambda = -1$. 1 / 1 point

   - ◉ True
   - ○ False

   Nice work

5. **True** or **False**: The Shifted Inverse Power Method for an invertible $n \times n$ matrix A is implemented by applying the Power Method to $A^{-1} - cI_n$ for some scalar c. 1 / 1 point

   - ○ True
   - ◉ False

   Nice work